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PhiliFinds
Software Requirements Specification
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Software Requirements Specification

1. Introduction

Travel planning can be overwhelming, especially when balancing budget, time constraints, transportation options, entrance fees, and accommodation arrangements. Many travelers spend hours comparing information across multiple platforms, which often leads to “planning fatigue” and inefficient itineraries.

PhiliFinds is an AI-powered travel planning platform design-ed to simplify and optimize the travel experience within the Philippines. The system provides personalized, data-driven itineraries tailored to users’ budgets, preferences, group size, and travel duration. By automating cost calculations, commute estimations, and schedule optimization, PhiliFinds helps travelers make informed decisions quickly and efficiently.

The platform serves as a comprehensive travel assistant that consolidates essential trip-planning components—including destinations, transportation, lodging, and estimated expenses—into a single, user-friendly system. Through intelligent recommendations and streamlined planning tools, PhiliFinds enables users to focus less on logistics and more on enjoying their travel experience.

1.1 Purpose

PhiliFinds aims to eliminate "planning fatigue" for travelers in the Philippines. By leveraging AI, the platform provides personalized, data-driven itineraries that optimize budget, travel time, and user preferences. It serves as a comprehensive travel assistant that handles the math of entrance fees, commute times, and group-sized lodging so the user can focus on the experience.

1.2 Scope

PhiliFinds is a progressive web application that focuses on:

- **AI Itinerary Helper:** Generates personalized travel itineraries based on user preferences, budget, duration, group size, and activity level.
- **Budget Optimization and Cost Estimation:** Calculating total projected trip costs including transit and fees.
- **Time and Logistics Management:** Optimizes travel schedules between destinations while considering group size and time constraints.
- **Navigation Assistance:** Provides point-to-point travel guidance and public transportation suggestions.
- **Progress Checklist:** Enables users to track completed and pending itinerary activities.
- **User Profile Management:** Stores user information, saved trips, past itineraries, and

emergency contact details.

- **Database Integration:** Uses Supabase for secure storage of user accounts and itinerary data.
- **Itinerary Sharing:** The system allows users to share itineraries with other users through a QR code or shareable link.
- **Notifications and Emergency Support:** Sends reminders for scheduled activities and displays emergency hotlines.
- **Out of Scope:** The system does not support direct booking of flights or accommodations and does not provide offline GPS navigation.

1.3 Definitions, Acronyms, and Abbreviations

PWA	Also known as a progressive web app, is an app that's built using web platform technologies, but that provides a user experience like that of a platform-specific app like mobile.
PhiliFinds	A progressive web app that utilizes AI for your travel itineraries.
AI	Artificial intelligence that will help the user pick the best deals for your travels.
Supabase	A cloud database for our PWA.
QR Code	A scannable code used to share itineraries with other users.

1.4 References

- **Next.js Documentation:** For frontend architecture and Server-Side Rendering (SSR).
- **Supabase Documentation:** For backend services, including Authentication and Row-Level Security (RLS).
- **Vercel Deployment Guide:** For ensuring 24/7 availability and global distribution.
- **Data Privacy Act of 2012 (Republic Act No. 10173):** To ensure all user data encryption and handling complies with Philippine law.
- **OpenAI/Anthropic API Documentation:** For implementing the core AI itinerary logic.

1.5 Overview

- Introduction: Defines the purpose, scope, and basic terminology of PhiliFinds.
- Overall Description**:** Provides a high-level view of the product's perspective, user classes, and operating environment.
- Specific Requirements: Details the functional and non-functional requirements, including performance benchmarks and security protocols necessary for successful deployment.

2. Overall Description

PhiliFinds is an independent Progressive Web Application (PWA) designed to bridge the gap

between complex travel logistics and user convenience.

- **Product Perspective:** The app operates as a cloud-based assistant, utilizing Supabase for real-time data storage and Next.js for high-performance delivery.
- **User Characteristics:** The platform caters to three main groups: Casual Tourists seeking ease of use, Business Travelers requiring efficient schedules, and Group Travelers needing cost-scaling for multiple people.
- **Assumptions and Dependencies:** It is assumed that users have an active internet connection to access AI services.

3. Specific Requirements

PhiliFinds is an AI-powered travel planning platform designed to simplify and optimize the travel experience within the Philippines by generating personalized, data-driven itineraries. The system provides a secure access gateway through account creation and user authentication, utilizing Supabase to store user credentials, profile information, emergency contact details, and saved itineraries. The authentication mechanism distinguishes between administrator and standard user roles and enforces a three-attempt login limit with a cooldown period to enhance security.

Once authenticated, users can access the Dynamic Itinerary Builder, which functions as an intelligent questionnaire that gathers essential travel inputs, including number of days of stay, preferred tour categories (historical, food trip, nature, museum, social/night life), activity intensity level, group size, starting point, end point, and travel budget. Users may also prioritize selected tourist destinations before itinerary generation.

The collected data is processed by the AI engine to generate an optimized itinerary that balances time constraints, destination proximity, travel duration, and group requirements. The system includes a Cost Estimator that provides real-time financial transparency by calculating entrance fees, ecological fees, activity fees, estimated meal expenses, and transportation costs based on user-defined budget limitations.

To enhance trip organization, the system includes a Time and Logistics Management component that optimizes scheduling between destinations and minimizes travel time. Navigation assistance is provided for point-to-point travel, including recommendations for public transportation options for users without private vehicles.

Users may modify AI-generated itineraries before finalization. Once satisfied, they may lock the itinerary, save it, print it, or share it via a unique URL or QR code. The system also includes a progress checklist feature that allows users to mark completed activities and track their travel progress.

Additionally, PhiliFinds provides a notification and reminder system for upcoming scheduled activities, as well as an emergency support interface that displays relevant hotlines and enables quick dialing on supported devices.

The application is built using Next.js to ensure fast initial page loads and mobile responsiveness. The AI engine is designed to generate optimized travel suggestions within 20 seconds to maintain high performance and usability standards.

Overall, PhiliFinds aims to eliminate “planning fatigue” by integrating itinerary generation, cost estimation, schedule optimization, navigation guidance, and trip management tools into a single secure and user-friendly platform, allowing travelers to focus more on their experience rather than logistical coordination.

3.1 Functionality

- Account Creation
- User Authentication
- Dynamic Itinerary Builder
- Cost Estimator
- Itinerary Sharing
- Progress Checklist
- Navigation Assistance
- Notifications and Reminders
- Emergency Support

PhiliFinds is an AI-powered travel planning platform that simplifies itinerary creation, cost estimation, and trip management for travelers in the Philippines. Users can securely create accounts, log in, and access a dynamic itinerary builder that collects travel preferences. The system generates optimized itineraries, calculates total costs, assists with navigation, provides reminders, supports emergency access, and enables sharing and tracking of travel plans.

3.1.1 User Account and Profile Management

3.1.1.1 The system shall ask for necessary details to complete account creation.

3.1.1.2 The system shall securely store registered account information in the database.

3.1.1.3 The system shall allow users to edit profile information.

3.1.1.4 The system shall store and display saved and past itineraries in the user profile.

3.1.2 User Authentication

3.1.2.1 The system will let the user login. If not matched, it will give 3 tries before cooldown.

3.1.2.2 The system will also determine if the account is an admin or a user.

3.1.3 Dynamic Itinerary Builder

3.1.3.1 The system shall collect travel inputs: destinations, duration, group size, budget, activity intensity, and preferences.

3.1.3.2 The system allows users to prioritize selected destinations before itinerary generation.

3.1.3.3 The system generates an AI-optimized itinerary based on time, proximity, and group requirements.

3.1.4 Cost Estimator

3.1.4.1 Calculates entrance fees, activity fees, ecological fees, meals, and transportation costs.

3.1.4.2 Adjusts calculations according to the user-defined budget.

3.1.4.3 Provides real-time cost transparency before itinerary finalization.

3.1.5 Itinerary Sharing

Allows users to share itineraries via a unique URL or QR code. Shared itineraries are read-only for other users.

3.1.6 Progress Checklist

Provides users with a checklist to mark completed activities and track pending itinerary tasks for better trip organization.

3.1.7 Navigation Assistance

Offers point-to-point directions between destinations and provides public transportation suggestions for users without private vehicles.

3.1.8 Notifications and Reminders

Sends reminders for upcoming itinerary activities to ensure timely participation and travel management.

3.1.9 Emergency Support

Displays relevant emergency hotline numbers and allows quick dialing on supported devices.

3.2 Usability

3.2.1 Mobile Responsive Design

The application shall be fully responsive and accessible across various screen sizes, including smartphones, tablets, laptops, and desktop devices, to ensure maximum device compatibility and usability.

3.2.2 Intuitive Onboarding

The application shall provide a clear and intuitive user interface that enables new users to navigate the system without prior training.

Users shall be able to complete the Dynamic Itinerary Builder process within **three (3) minutes** under normal network conditions.

3.2.3 Accessibility

The application shall ensure:

- Fast content rendering and minimal loading delays.
- Clear typography and appropriate font sizing.
- Sufficient color contrast for readability.
- Logical navigation structure for ease of use on both desktop and mobile devices.

3.3 Reliability & Availability

3.3.1 Application Availability

The system shall maintain high availability and operate 24/7 under normal operating conditions.

Deployment through Supabase (backend services) and Vercel (hosting platform) shall support continuous operation and minimize service downtime.

3.4 Performance

The application shall provide fast initial page loads through optimized frontend architecture using Next.js.
The system shall:

- Minimize loading times for itinerary generation and navigation between pages.
- Maintain smooth performance on mobile devices with limited hardware capabilities.
- Generate AI-based itinerary suggestions within **20 seconds** under normal network conditions.

3.5 Security

PhiliFinds shall implement appropriate security measures to protect user data, system integrity, and platform availability. The system shall ensure confidentiality, integrity, and availability of user information at all times.

3.5.1 Row-Level Security

The system shall implement Supabase Row-Level Security (RLS) policies to ensure that users can only view, edit, and manage their own itineraries and personal data.

3.5.2 Data Encryption

All sensitive user data shall be encrypted in transit using HTTPS/TLS protocols.

Stored data shall be protected using secure database encryption mechanisms provided by the backend infrastructure.

3.5.3 Secure API Keys

AI and map service API keys shall be stored in secure environment variables and shall not be exposed to the client-side application.

3.6 Design Constraints

3.6.1 Platform Constraints

As a Progressive Web Application (PWA), the system shall:

- Be compatible with modern web browsers such as Chrome, Safari, and Edge.
- Maintain high performance across devices with varying hardware capabilities.
- Operate primarily in an online environment with active internet connectivity.

3.6.2 Data Constraints

The system shall adhere to the storage limitations of the Supabase free tier.

AI-generated data and itinerary processing shall remain within defined performance thresholds to prevent excessive resource consumption.

3.7 Interfaces

3.7.1 User Interface

PhiliFinds shall provide a clean, intuitive PWA-based interface developed using Next.js.

The system shall include the following primary screens:

- **Home Page:** Provides quick access to itinerary creation, saved trips, and user profile.
- **Login and Registration Page:** Secure forms for account creation and authentication via Supabase.
- **Itinerary Builder Page:** Interactive questionnaire for destination, travel duration, group size, preferences, and budget inputs.
- **Results Page:** Displays AI-generated itineraries, estimated costs, and location details.
- **Profile/Settings Page:** Allows users to view, edit, or delete their account and manage saved itineraries.

The user interface shall follow responsive design principles and comply with accessibility standards for readability and navigation on both desktop and mobile platforms.

The UI must follow **responsive design** principles and meet accessibility standards for readability and navigation on both desktop and mobile devices.

3.7.2 Hardware Interface

PhiliFinds is a Progressive Web Application accessible through any device equipped with:

- A modern web browser
- An active internet connection

Supported hardware includes:

- Smartphones (Android and iOS)
- Tablets
- Laptops and Desktop Computers

No specialized hardware components are required. Optional device features such as location services may be utilized for map and navigation functionalities.

3.7.3 Software Interface

PhiliFinds shall integrate with the following external systems and APIs:

- **Supabase API:** Handles authentication, user profile management, itinerary storage, and Row-Level Security (RLS).
- **OpenAI/Anthropic API:** Generates AI-based travel recommendations and itinerary optimization.
- **Map/Geolocation API (e.g., Google Maps or OpenStreetMap):** Provides travel routes, estimated distances, and time calculations.
- **Vercel Platform:** Manages application deployment, scalability, and hosting infrastructure.

All API keys and database credentials shall be securely stored in environment variables and shall not be exposed to client-side code.